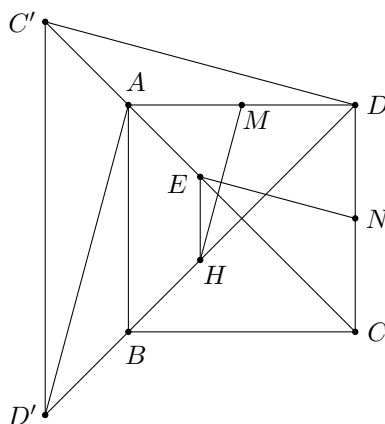




HK IMO Training 2023-2024 Problem Set 14 Suggested Solutions



Problem 53 (Tournament of Towns 2022 Spring Junior O-Level Problem 4). In a square $ABCD$, let E be a point on the diagonal AC . Let H be the orthocentre of $\triangle ADE$, and let M and N be the midpoints of AD and CD respectively. Prove that $HM \perp EN$.



Solution. Since $AE \perp DH$ and $AC \perp BD$, H lies on BD . Let C' be the symmetric point of C about E , and let D' be the symmetric point of D about H . By the midpoint theorem, we have $HM \parallel D'A$ and $EN \parallel C'D$. Therefore, it suffices to prove $D'A \perp C'D$.

As $EH \parallel DC$, we have $C'D' \parallel DC$. This implies $C'D' \perp AD$. Also, it is clear that $DD' \perp AC'$. Therefore, A is the orthocentre of $\triangle C'D'D$. It follows that $C'D \perp AD'$ as desired. $\square$

Problem 54 (Tournament of Towns 2021 Spring Senior A-Level Problem 2 modified). Find all positive integers n with the following property: For any real numbers x and y , there exist real numbers $a_1, a_2, \dots, a_n$ such that

$$x = a_1 + a_2 + \dots + a_n \quad \text{and} \quad y = \frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}.$$

Solution. Answer: All integers $n \geq 4$.

If the property is satisfied for some positive integer $n = m$, then the property is satisfied for all $n \geq m$. This is because for any real numbers x and y , we can set $a_{m+1} = a_{m+2} = \dots = a_n = 1$, and solve

$$x - (n - m) = a_1 + a_2 + \dots + a_m \quad \text{and} \quad y - (n - m) = \frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_m}.$$

Therefore, it suffices to show that the property is not satisfied when $n = 3$, and is satisfied when $n = 4$.

When $n = 3$, we show that there is no solution if $x = y = 0$. Indeed, we need $a_3 = -(a_1 + a_2)$ from the first equation. Then the second equation becomes $\frac{1}{a_1} + \frac{1}{a_2} = \frac{1}{a_1 + a_2}$, of $a_1^2 + a_1a_2 + a_2^2 = 0$. There is no nonzero solution.

When $n = 4$, given any $x, y \in \mathbb{R}$, we can always find $a_4 \in \mathbb{R}$ such that $x' = x - a_4$ and $y' = y - \frac{1}{a_4}$ are nonzero. It remains to solve

$$x' = a_1 + a_2 + a_3 \quad \text{and} \quad y' = \frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3}.$$

Assuming a_3 is fixed, these equations are equivalent to

$$a_1 + a_2 = x' - a_3 \quad \text{and} \quad a_1a_2 = \frac{a_1 + a_2}{y' - a_3^{-1}} = \frac{x' - a_3}{y' - a_3^{-1}}.$$

This has a nonzero solution iff $(x' - a_3)^2 \geq \frac{4(x' - a_3)}{y' - a_3^{-1}}$ and $x' - a_3 \neq 0$. Consider

$$f(a_3) = (x' - a_3)^2 - \frac{4(x' - a_3)}{y' - a_3^{-1}}.$$

When $a_3 \rightarrow \infty$, the dominating term is a_3^2 , so $f(a_3) \rightarrow \infty$. Thus, there exists a sufficiently large a_3 such that $f(a_3) > 0$ and $x' - a_3 < 0$. For this value of a_3 , we can solve a_1 and a_2 uniquely. This completes the proof. $\square$

Problem 55 (USAMO 2019 Problem 5). Initially, there are two numbers r and $\frac{1}{r}$ written on the blackboard where r is a positive rational number. Every time we can choose two numbers on the blackboard and write their arithmetic mean or their harmonic mean on the blackboard. Find all r such that the number 1 can be written on the blackboard in finitely many steps.

Solution. Answer: $r = \frac{m}{n}$ where $(m, n) = 1$ and $m + n$ is a power of 2.

Let $r = \frac{m}{n}$ where $(m, n) = 1$. We first show that $m + n$ has to be a power of 2. If not, we pick any odd prime p dividing $m + n$. We claim that p divides the sum of the numerator and the denominator of each number that may appear on the blackboard. Indeed, initially, both $r = \frac{m}{n}$ and $\frac{1}{r} = \frac{n}{m}$ have this property. Suppose $\frac{a}{b}$ and $\frac{c}{d}$ are two fractions on the blackboard with $(a, b) = (c, d) = 1$, $p \mid a + b$ and $p \mid c + d$. Note that $p \nmid b$ (otherwise $p \mid a$ and $p \mid b$ from $p \mid a + b$, contradicting $(a, b) = 1$) and $p \nmid d$. Therefore, we can write $\frac{a}{b} \equiv \frac{a}{-a} \equiv -1 \pmod{p}$ and $\frac{c}{d} \equiv -1 \pmod{p}$. We check that

$$\begin{aligned} \frac{1}{2} \left(\frac{a}{b} + \frac{c}{d} \right) &\equiv \frac{1}{2}(-1 - 1) \equiv -1 \pmod{p}, \\ \frac{\frac{a}{b} + \frac{c}{d}}{2} &\equiv \frac{-1 - 1}{-1 - 1} \equiv -1 \pmod{p}. \end{aligned}$$

In other words, all new numbers written on the blackboard are congruent to -1 modulo p . If a new number is $\frac{a}{b}$, then we must have $a \equiv -b \pmod{p}$, or $p \mid a + b$. As $p \nmid 1 + 1$, the number $1 = \frac{1}{1}$ cannot be written on the blackboard.

If $m + n = 2^k$ for some positive integer k , we can always write down the number 1 by using only arithmetic means. Given any numbers s and t on the blackboard, we first show that we can obtain any weighted mean $\frac{as + bt}{a + b}$ where a and b are odd positive integers and $a + b = 2^\ell$ for some $\ell \geq 1$. This is obvious if $\ell = 1$, as the only such number is $\frac{s + t}{2}$. Suppose this holds for some ℓ . Consider $a + b = 2^{\ell+1}$ where a and b are odd positive integers. WLOG assume $a < 2^\ell < b$. Then

$$\frac{as + bt}{2^{\ell+1}} = \frac{1}{2} \left(\frac{as + (b - 2^\ell)t}{2^\ell} + t \right).$$

This means we can take the average of $\frac{as + (b - 2^\ell)t}{2^\ell}$ and t to obtain $\frac{as + bt}{2^{\ell+1}}$. By induction on ℓ , the claim follows. In particular, we can generate the number

$$\frac{n \cdot \frac{m}{n} + m \cdot \frac{n}{m}}{2^k} = \frac{m + n}{2^k} = 1$$

by using only arithmetic means. $\square$

Problem 56 (ELMO 2020 Problem 6). There is a calculator which displays a certain integer $a > 1$ initially. The calculator has three buttons. By pressing these three buttons respectively, we can change the current displayed number n to $d(n)$, $\sigma(n)$ and $\varphi(n)$ respectively. Prove that we can make the calculator display any integer $b > 1$ after pressing the buttons finitely many times. (Here, $d(n)$ is the number of positive divisors of n , $\sigma(n)$ is the sum of all positive divisors of n , and $\varphi(n)$ is the number of integers from 1 to n which are relatively prime to n .)

Solution. Let $\Omega(n)$ be the number of prime divisors of n counted with multiplicity.

Claim 1. $\Omega(\varphi(n)) \geq \Omega(n) - 1$ for all n , and equality holds iff n is of the form $2^\alpha 3^\beta$ with $\alpha \geq 1$.

Proof. Let $n = p_1^{\alpha_1} \cdots p_s^{\alpha_s}$ be the prime factorization where $p_1 < \cdots < p_s$. Then

$$\varphi(n) = p_1^{\alpha_1-1} \cdots p_s^{\alpha_s-1} (p_1 - 1) \cdots (p_s - 1).$$

This implies $\Omega(\varphi(n)) - \Omega(n) = \Omega(p_1 - 1) + \cdots + \Omega(p_s - 1) - s$. This is nonnegative unless $p_1 = 2$ and $p_2 - 1, \dots, p_s - 1$ are all primes. But since $p_2, \dots, p_s$ are primes, the only possibility is $s = 1$, or $s = 2$ and $p_2 = 3$. For $\alpha, \beta \geq 1$, we check that

$$\begin{aligned} \Omega(\varphi(1)) &= \Omega(1), \\ \Omega(\varphi(2^\alpha)) &= \Omega(2^{\alpha-1}) = \alpha - 1 = \Omega(2^\alpha) - 1, \\ \Omega(\varphi(3^\beta)) &= \Omega(2 \cdot 3^{\beta-1}) = \beta = \Omega(3^\beta), \\ \Omega(\varphi(2^\alpha 3^\beta)) &= \Omega(2^\alpha 3^{\beta-1}) = \alpha + \beta - 1 = \Omega(2^\alpha 3^\beta) - 1. \end{aligned}$$

$\square$

Claim 2. $v_2(\varphi(n)) \geq v_2(n)$ for all n except those of the form 2^α with $\alpha \geq 1$.

Proof. Let $n = 2^\alpha m$ where m is odd. If $\alpha = 0$ and $m \geq 3$, then $\varphi(n) = \varphi(m)$ is even, and so $v_2(\varphi(n)) \geq 1 > v_2(n)$. If $\alpha \geq 1$ and $m \geq 3$, then $v_2(\varphi(n)) = v_2(2^{\alpha-1}\varphi(m)) \geq \alpha = v_2(n)$. Finally, when $m = 1$, we check that

$$\begin{aligned} v_2(\varphi(2^\alpha)) &= v_2(2^{\alpha-1}) = \alpha - 1 < v_2(2^\alpha) \quad \text{for } \alpha \geq 1, \\ v_2(\varphi(1)) &= v_2(1). \end{aligned}$$

□

For every prime p , note that $\varphi(p), \varphi^2(p), \dots$ is a strictly decreasing sequence of even positive integers until it decreases to 2 and then 1. Let $f(p)$ be the first term in this sequence which is equal to 2 times a composite number or equal to 2.

Claim 3. For any integer C , we have $f(p) > C$ for all sufficiently large prime p .

Proof. If this is not true, then there exists C such that $f(p) \leq C$ for infinitely many primes p . By the pigeonhole principle, there exists k and infinitely many primes p such that $f(p) = k$. We may assume $k > 2$ since if $k = 2$, the term before 2 can only be 4, 6 (except $p = 3$), and we can replace k by 4 or 6. By definition, all terms before k are 2 times a prime. As $\varphi(2q) = q - 1$ for prime $q \geq 3$, the terms before k must be $2(k+1), 2(2k+3), 2(4k+7), \dots, 2(2^j k + 2^{j+1} - 1), \dots$. However, note that

$$2^{\varphi(k+1)}k + 2^{\varphi(k+1)+1} - 1 \equiv -2^{\varphi(k+1)} + 2^{\varphi(k+1)+1} - 1 = 2^{\varphi(k+1)} - 1 \equiv 0 \pmod{k+1}$$

as k is even. This shows $2^{\varphi(k+1)}k + 2^{\varphi(k+1)+1} - 1$ cannot be a prime. Thus, the sequence cannot be extended indefinitely, which means there cannot be infinitely many primes p with $f(p) = k$. □

Claim 4. For any integer C , and for all sufficiently large prime p , we have $\Omega(\varphi^j(p)) \geq C$ for some $j \geq 1$.

Proof. We prove this by induction on C . The base case $C = 1$ is trivial. Assume for all primes $p \geq N \geq 2$, we have $\Omega(\varphi^j(p)) \geq C$ for some $j \geq 1$. We want to prove the statement for the case $C + 1$. For all sufficiently large prime p , we have $f(p) > N^{C+1}$ by Claim 3.

- If all prime divisors of $f(p)$ are at most N , then $\Omega(f(p)) \geq C + 1$. This implies $\Omega(\varphi^j(p)) \geq C + 1$ for some j .
- If $f(p)$ has a prime divisor $q > N$, then $f(p) = 2qm$ for some $m > 1$. By definition, we have $\Omega(\varphi^j(q)) \geq C$ for some $j \geq 1$. Thus, there exists $\ell \geq j$ such that $\Omega(\varphi^\ell(p)) = \Omega(\varphi^\ell(2qm)) \geq \Omega(\varphi^j(q)) \geq C$ as $\varphi^j(q) \mid \varphi^\ell(2qm)$. It remains to show that the equality cannot hold.

Indeed, to have $\Omega(\varphi^j(2qm)) = \Omega(\varphi^j(q))$, we need $\varphi^j(q) = \varphi^j(2qm)$. This cannot hold when $j = 1$ since $m > 1$. When $j \geq 2$, as $\varphi(n)$ is even for $n > 1$, we only need to show that $\varphi(2x) \neq \varphi(2y)$ if $x \mid y$ and $x \neq y$. Indeed, if $x = 2^{\alpha_1} p_2^{\alpha_2} \dots p_s^{\alpha_s}$ and $y = 2^{\beta_1} p_2^{\beta_2} \dots p_s^{\beta_s} p_{s+1}^{\beta_{s+1}} \dots p_t^{\beta_t}$, then

$$\varphi(2x) = 2^{\alpha_1} p_2^{\alpha_2-1} \dots p_s^{\alpha_s-1} (p_2 - 1) \dots (p_s - 1) < 2^{\beta_1} p_2^{\beta_2-1} \dots p_t^{\beta_t-1} (p_2 - 1) \dots (p_t - 1) = \varphi(2y).$$

□

Claim 5. For any integer C , and for all sufficiently large integer n , we have $\Omega(\varphi^j(n)) \geq C$ for some $j \geq 1$.

Proof. If n is sufficiently large, either it has a large prime divisor p or it has at least $C + 1$ prime divisors. In the former case, we are done by Claim 4 as $\varphi^j(p) \mid \varphi^j(n)$, and so $\Omega(\varphi^j(n)) \geq \Omega(\varphi^j(p)) \geq C$. In the latter case, we have $\Omega(\varphi(n)) \geq \Omega(n) - 1 \geq C$ by Claim 1. □

We can now solve the original problem. Since $\sigma(n) > n$ for $n \geq 2$, we can press σ many times until the displayed number n is sufficiently large. By Claim 5, we can have $\Omega(\varphi^j(n)) \geq 2^b + b$ for some j . We can press φ many times until the displayed number becomes $2^\alpha 3^\beta$ (which must happen as the number eventually becomes 2). As the number of prime divisors decreases afterwards, we must have $\alpha + \beta \geq 2^b + b$ at this moment by the assumption.

- If $\alpha \geq b$, we continue to press φ until the displayed number becomes 2^{b-1} (note that $\varphi(2^x 3^y) = 2^{x-1} 3^y$ for $y \geq 1$ and $\varphi(2^x) = 2^{x-1}$). Then we press d to obtain $d(2^{b-1}) = b$.
- If $\beta \geq 2^b$, we continue to press φ until the displayed number becomes $2^\alpha 3^{2^b-1}$. Afterwards, we press d to obtain $(\alpha + 1)2^b$. We then press φ many times until the displayed number is of the form 2^γ for the first time. By Claim 2, we must have $\gamma = v_2(2^\gamma) \geq v_2((\alpha + 1)2^b) \geq b$. Thus, we can continue to press φ many times until the displayed number becomes 2^{b-1} . We are done by pressing d .

□